

Query Designer Capabilities

- **Selecting Entities and Attributes**
- **Filters**
- **Sorting**
- **Alias**
- **Joins**
- **Aggregation**
- **Expression Attributes**
- **Alternative/Non-base Views**
- **Time zone Selection**

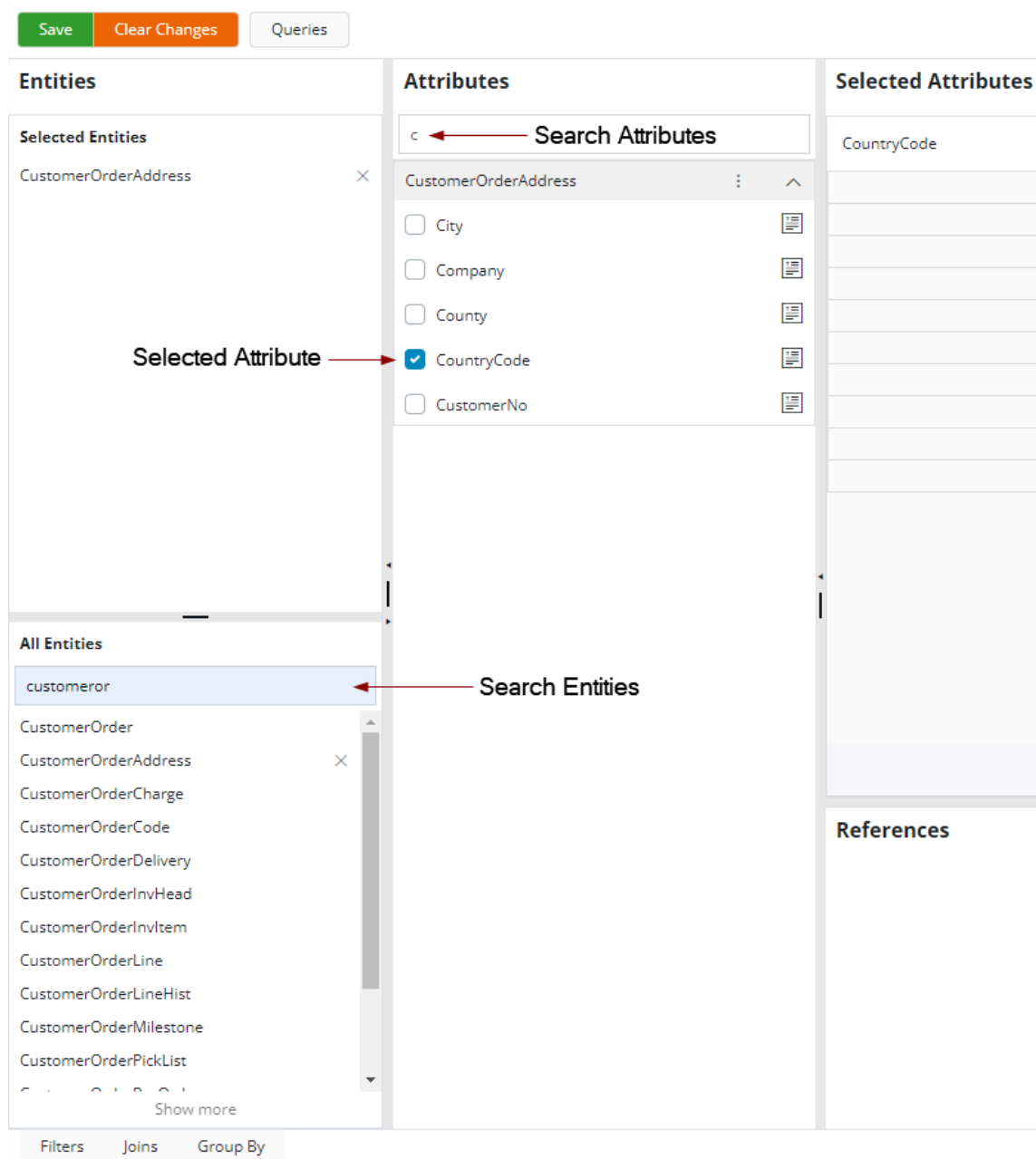
Selecting Entities and Attributes

Adding an entity to the query can be done through the entities Pane on the Query Designer. To select an entity simply click on the entity name listed in the *All Entities* pane.

To search an entity, type in the search phrase on *Search entities* text field and press enter. This will search the first 25 entities where the entity name starts with the search phrase. If the entity you search for is not listed within the search result try clicking on *Show more* button at the bottom of the list.

After selecting an entity, the Query Designer will display all the attributes of the entity in the *Attributes* pane. Expand the relevant entity to see the attributes by clicking on ▼ icon next to the entity name. The data type of each attribute is indicated right next to the attribute name. To select an attribute click on the attribute name or the checkbox. *Selected Attributes* are indicated by the checkmark to the left of the attribute name.

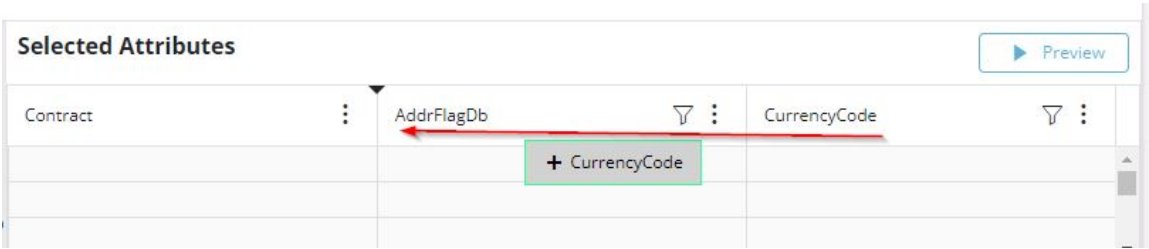
Users also can search for attributes by typing the attribute name in *Search attributes* text field.



Selected Attributes can be deselected by clicking on the attribute name. The checkmark will be removed indicating the attribute is no longer used in the query.

To remove a *selected* entity, find the entity either in the *Selected Entities* pane or the *All Entities* pane and click on the [X] button next to the entity name. This action will remove every information related to the entity from the query.

Users can reorder the *Selected Attributes* by dragging and dropping them to the new location within the *Selected Attributes* pane.



Filters

Basic Filters

A basic filter can be applied to any *Selected Attributes*. Click on the menu button of the attribute in the *Selected Attributes* pane. Select *Filter* menu. Based on the datatype of the attribute a list of supported filter conditions and an input field will be presented. Select the required condition and the value. Click on Apply. The filter is now applied to the attribute and it will be indicated by an icon next to the attribute name in the *Selected Attributes* pane.

ZipCode

⌵ ⋮

↑ Sort Ascending

↓ Sort Descending

⌵ Filter

is equal to

80000

Clear

Apply

AdditionalDiscount

⋮

AdditionalDiscount

⋮

DefaultCharges

⋮

↑ Sort Ascending

↓ Sort Descending

⌵ Filter

is equal to

Clear

Apply

Filter menu for Text Attribute

Filter menu for Numerical Attribute

ZipCode

AdditionalDiscount

Sort Ascending
Sort Descending
Filter

is equal to

80000

Clear

Apply

AdditionalDiscount

DefaultCharges

Sort Ascending
Sort Descending
Filter

is equal to

Clear

Apply

DefaultCharges

ExpectedPrepaymentDate

Sort Ascending
Sort Descending
Filter
Is True
Is False

Clear

Apply

ExpectedPrepaymentDate

Sort Ascending
Sort Descending
Filter

is equal to

10/16/2021

May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
2022
Feb
Mar

October 2021

TODAY

SU MO TU WE TH FR SA

1 2

3 4 5 6 7 8 9

10 11 12 13 14 15 16

17 18 19 20 21 22 23

24 25 26 27 28 29 30

31

Filter menu for Boolean Attribute

Filter menu for Date/Time Attribute

Basic filters can be applied to any number of *Selected Attributes* at the individual attribute level. Each condition will be joined with an **AND** operator.

Filter Pane

The filter pane allows the user to create advanced filters. To open/close the filter pane, click on the **Filter** button at the bottom of the Query Designer.

Save Clear Changes Queries

Entities

Selected Entities

CustomerOrder

All Entities

customer

CustomerOrder

CustomerOrderAddress

CustomerOrderCharge

CustomerOrderCode

CustomerOrderDelivery

Show more

Attributes

Search attributes

PayTermBaseDate

PrepaymentApprovedDb

CurrencyCode

DateEntered

DeliveryCountryCode

ApplyFixDelivFreight

DeliveryTerms

ForwardAgentId

Noteld

SmConnectionDb

GrpDiscCalcFlagDb

Physical addresses

Selected Attributes

AddFlag

Contract

CurrencyCode

(No data)

(Limited to 10 rows)

References

AND OR

CustomerOrder CurrencyCode is equal to USD

CustomerOrder CustomerNo is equal to 2

Add Condition Add Condition Group

To open/close the Filter pane

Filters Joins Group By

To create a filter using the Filter pane, click on the **Add Condition** or **Add Condition Group** button. The **Add Condition Group** button can be used to create more complex filters by creating nested filter conditions within another Condition Group.

AND OR

IncomeType Description is equal to Parameter DescriptionParam

IncomeType CurrencyCode is equal to Value

Add Condition Add Condition Group

The four inputs in a filter condition are as follows,

1. The first dropdown box will list all the entities selected into the current query.
2. After selecting the entity from the first dropdown, the second dropdown will display all the attributes of the selected entity.
3. Based on the data type of the selected attribute the third dropdown will display a list of supported conditional operators
4. The fourth input box will be displayed after selecting the attribute. This dropdown will allow users to choose the type of filter condition, either "value" or "parameter".
5. The fifth input box will be displayed after selecting the attribute. This input box will be different based on the data type of the selected attribute, value type of the condition and the selected filter operator.

After selecting all four values for the dropdown of the filter condition, The user can either enter the value to be filtered into the input field or enter a parameter name.

When queries with parameters are exposed as a REST API, the parameters will be mapped to OData function parameters.

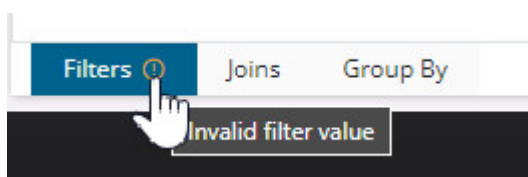
The screenshot shows the Filter pane in the Query Designer. At the top, there are radio buttons for 'AND' (selected) and 'OR'. Below this, there are two filter conditions. The first condition is 'IncomeType' is equal to 'Description' with parameter 'DescriptionParam'. The second condition is 'IncomeType' is equal to 'CurrencyCode' with parameter 'Parameter'. A dropdown menu is open for the second condition, showing a search bar and a list of parameters including 'DescriptionParam'. There are buttons for 'Add Condition' and 'Add Condition Group' at the bottom left of the pane.

To create a parameter type filter condition, the user can either create a new parameter or select an existing parameter. Once the user starts typing a variable name, the button "Add Parameter" will be enabled, and the existing parameter names of the selected attribute's datatype will be filtered based on the typed value.

To remove a filter condition, click on the [X] button next to the filter condition.

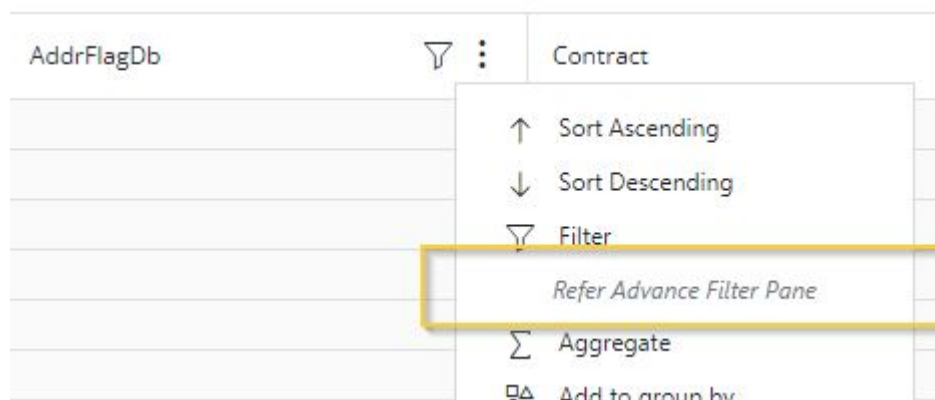
Users can select either logical **AND** or logical **OR** as the logical operator of the filters by clicking on the radio button on the top of the Filter pane.

The Query Designer will not allow the user to add more conditions or condition groups if there are incomplete filter conditions and, it will warn the user. These warnings will be visible on the **Filter** pane button on the bottom of the Query Designer even if the Filter pane is closed. To see more details of the warning, hover over the (i) icon.



Note:

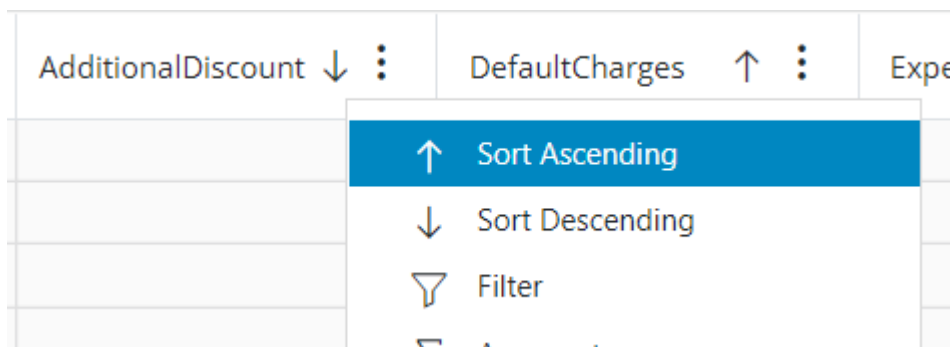
- Filter conditions created in the *Selected Attributes* pane are also displayed and can be modified in the Filter pane.
- If the filter conditions created in the Filter pane are too complex, the Basic Filters option will be disabled in the *Selected Attributes* pane.



- The query cannot be previewed or saved if Filter pane warnings exist. To preview or save the query, either remove or complete the erroneous filter condition.

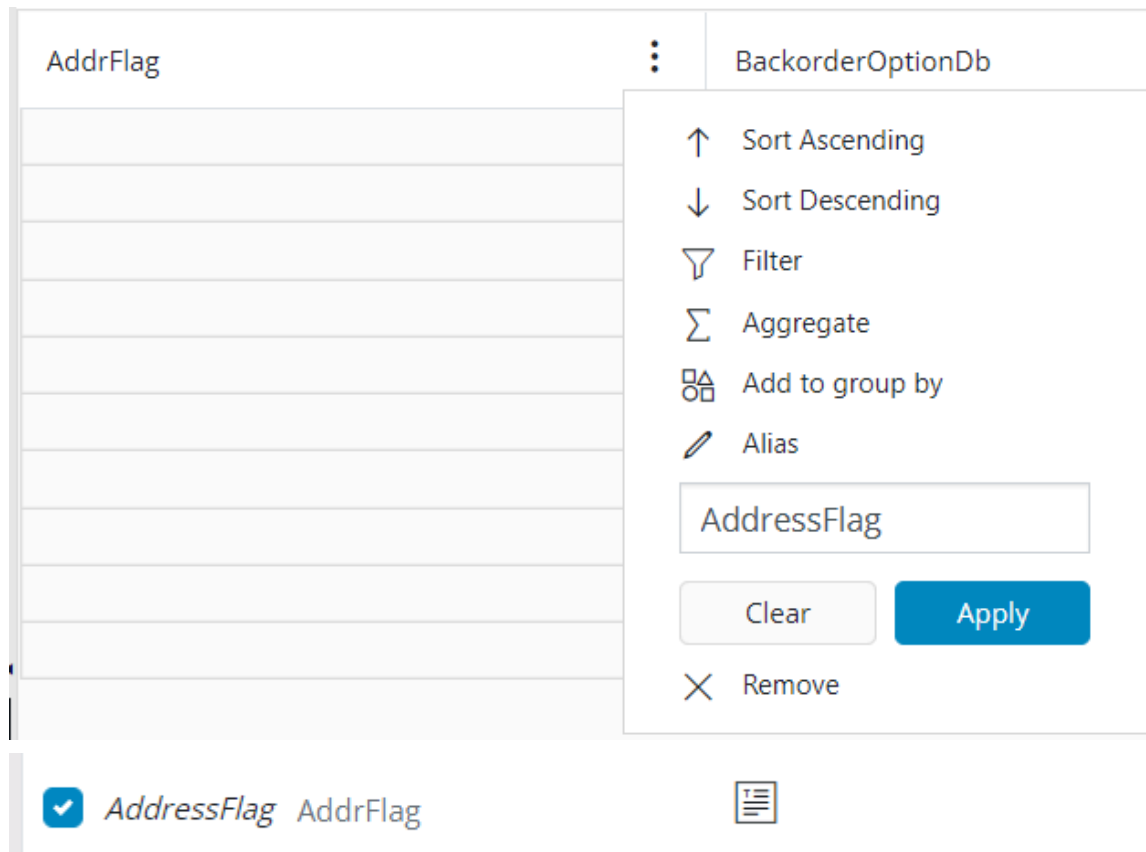
Sorting

The query can be sorted by one or more *Selected Attributes*. To do so, click on the menu button of the selected attribute in the *Selected Attributes* pane and select the sorting direction. Repeating the same will toggle the action. The sorting direction will be indicated on the *Attributes* pane next to the relevant attribute and the *Selected Attributes* pane.



Alias

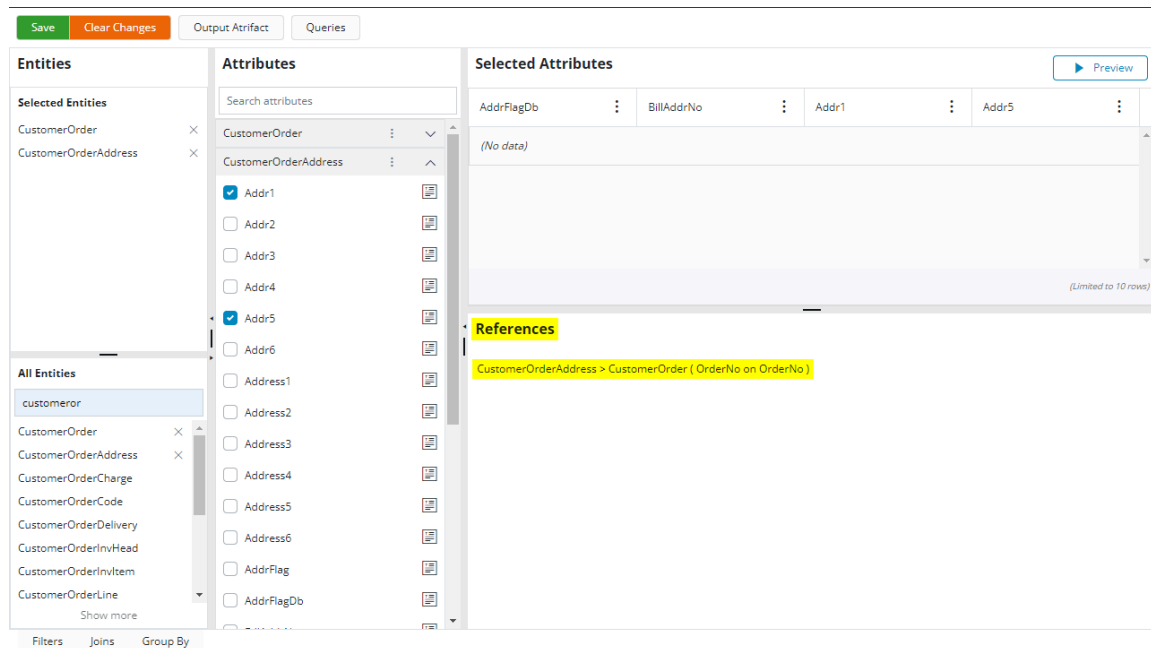
To set an alias to a selected attribute click on the menu button next to the relevant attribute's name in the *Selected Attributes* pane. Click on Alias. Type in the alias and click Apply. The alias will be displayed on the *Selected Attributes* pane as well as on the *Attributes* pane. The original attribute name will also be visible next to the alias.



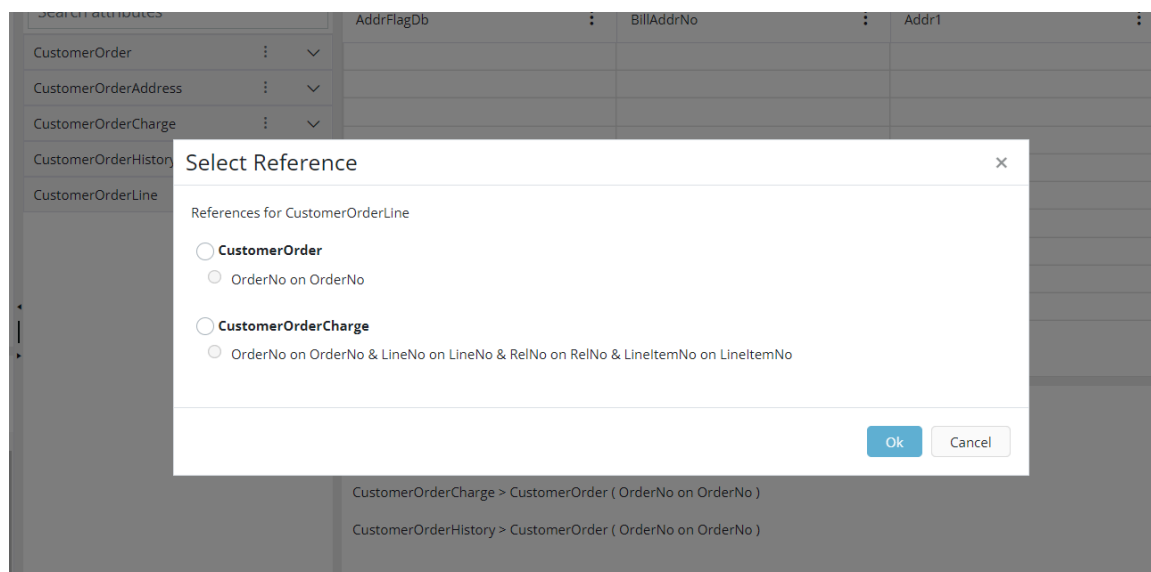
Note: Only letters, numbers and, underscore (_) are allowed

Joins

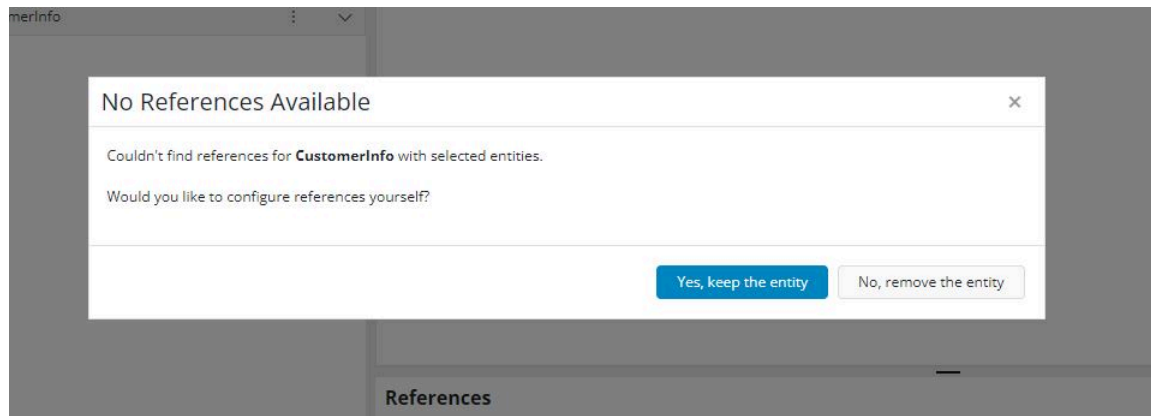
The Query Designer has the capability of joining entities automatically based on the reference details. When a user selects more than one entity, the Query Designer will look for all the reference details with previously selected entities and add them into the join details. The references being used will be displayed in the References pane.



In case there are multiple reference details found, the Query Designer will prompt the user to select the reference to be used in their query.



If the selected entity does not have any reference with previously selected entities, the Query Designer will prompt for the user confirmation to keep the entity or remove it.

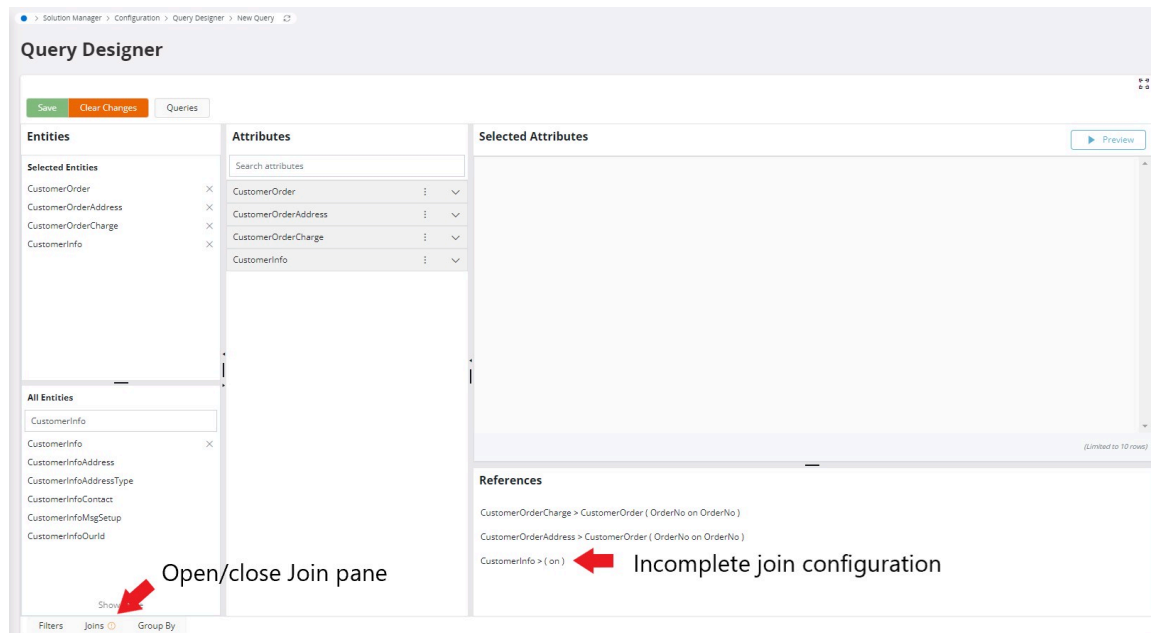


To add the entity into the Query Designer and manually configure the Join conditions, click on **Yes, keep the entity** button. Otherwise, click on **No, remove the entity** to remove it from the query.

Join Pane

The Join pane allows the user to modify the entity joining conditions. Moreover, it enables the user to add entities that do not have predefined reference information into the query.

In the Join pane, each row represents an individual join condition between two entities selected into the query. Users can modify these conditions as needed.



Save Clear Changes Output Atrifact Queries

Entities

Selected Entities

- CustomerOrder
- CustomerOrderAddress
- CustomerOrderCharge

All Entities

- CustomerInfo

Attributes

Search attributes

- CustomerOrder
- CustomerOrderAddress
- CustomerOrderCharge
- CustomerInfo

Selected Attributes

Preview

(Limited to 10 rows)

Join Type	Connections				
INNER	<table border="1"> <tr> <td>CustomerOrderCharge</td> <td>CustomerOrder</td> </tr> <tr> <td>OrderNo</td> <td>OrderNo</td> </tr> </table>	CustomerOrderCharge	CustomerOrder	OrderNo	OrderNo
CustomerOrderCharge	CustomerOrder				
OrderNo	OrderNo				
INNER	<table border="1"> <tr> <td>CustomerOrderAddress</td> <td>CustomerOrder</td> </tr> <tr> <td>OrderNo</td> <td>OrderNo</td> </tr> </table>	CustomerOrderAddress	CustomerOrder	OrderNo	OrderNo
CustomerOrderAddress	CustomerOrder				
OrderNo	OrderNo				
INNER	<table border="1"> <tr> <td>CustomerInfo</td> <td>CustomerOrder</td> </tr> <tr> <td></td> <td>OrderNo</td> </tr> </table>	CustomerInfo	CustomerOrder		OrderNo
CustomerInfo	CustomerOrder				
	OrderNo				

Filters Joins Group By

To configure an incomplete join configuration, the user must select an entity from the 2nd dropdown to join with (in this case) *CustomerInfo* entity. Then using the 3rd and 4th dropdowns, the user must select the attributes from each entity that should be used in the join configuration.

Join Type

INNER

Connections

CustomerOrderAddress	CustomerOrder
OrderNo	OrderNo
CustomerOrderCharge	CustomerOrder
OrderNo	OrderNo
CustomerInfo	CustomerOrder
CustomerNo	CustomerNo

Filters Joins

By default, the Query Designer creates **INNER** joins. To change this to a **LEFT OUTER** join, use the Join Type dropdowns.

Join Type

INNER

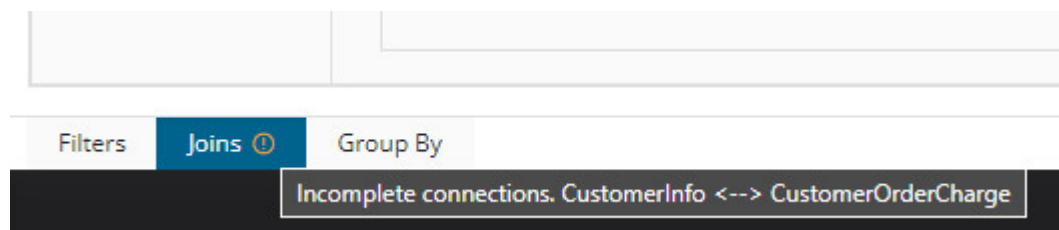
Connections

CustomerOrder	CustomerOrderCharge
OrderNo	OrderNo
CustomerOrderAddress	CustomerOrder
OrderNo	OrderNo
CountryCode	CountryCode
CustomerInfo	CustomerOrder
CustomerNo	CustomerNo

Filters Joins

Using the [+] button (1) next to the attributes dropdowns, the user can add more attribute combinations into the join configuration. To remove an attribute pair from a join configuration click on the [X] button (2) next to the relevant attributes pair.

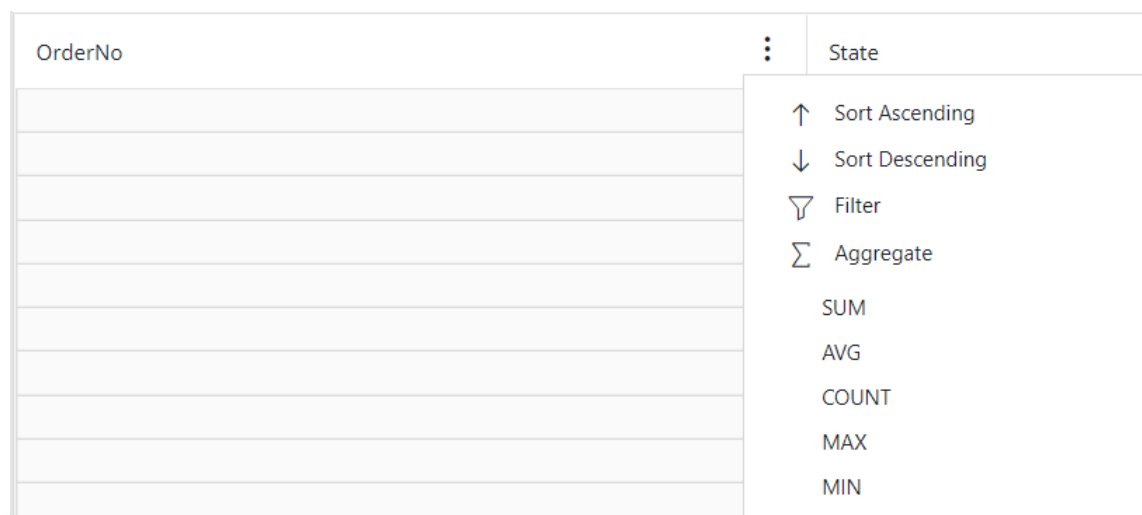
The Query Designer will warn the user if there are incomplete join configurations. The query cannot be previewed or saved if Join pane warnings exist. To preview or save the query, either remove or complete the erroneous join condition.



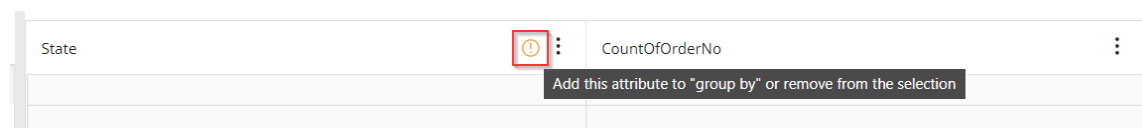
When an entity is removed from the query, join information related to the removing entity will also be removed from the query.

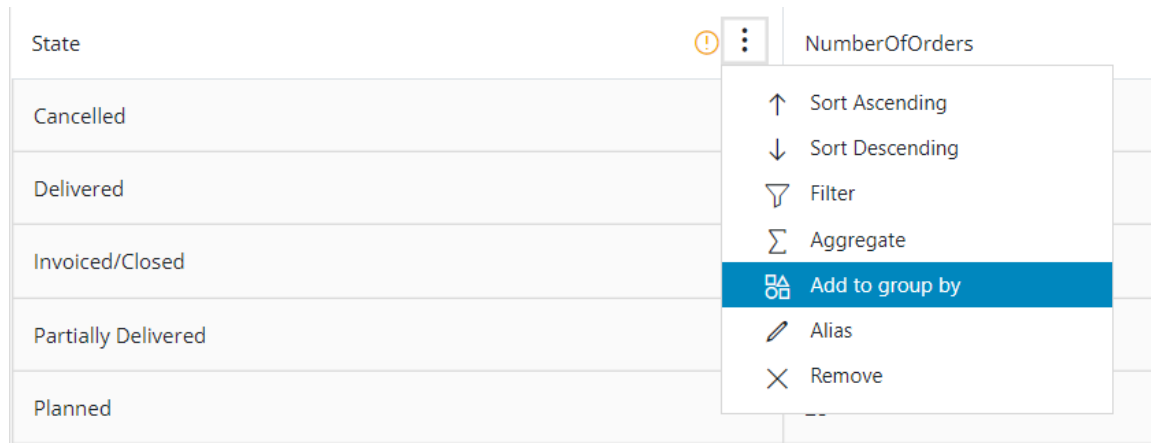
Aggregation

To apply aggregation to a selected attribute, click on the menu button next to the relevant attribute's name in the *Selected Attributes* pane. Click on *Aggregate* and select the required aggregation function.

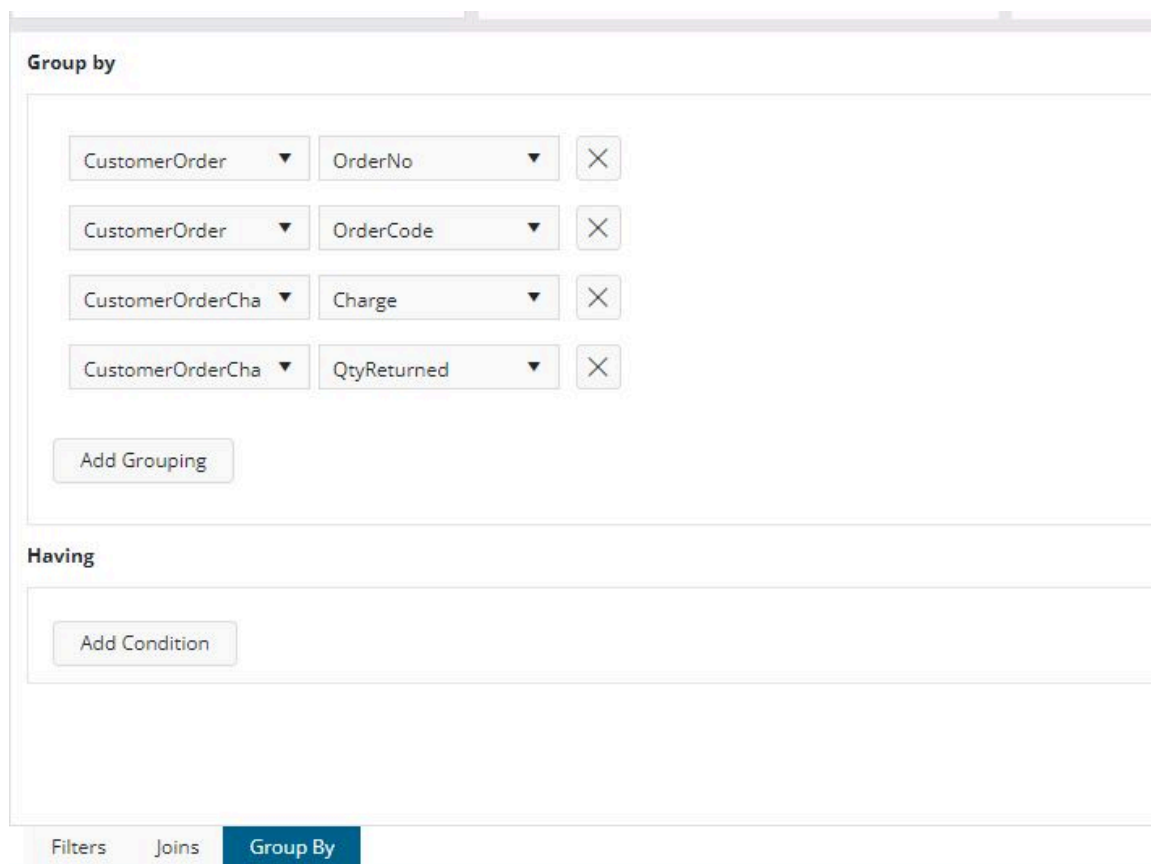


If the query has multiple attributes, the Query Designer will indicate that the query should be *grouped by* the other attributes. To do so, click on the menu button of the same attribute and click on *Add to group by*.





Users can further modify the attributes in Group By using the advance Group By pane.



To group the result by an attribute, click on the **Add Grouping** button. An empty grouping rule will be created. Fill in the rule by selecting the entity and the attribute of the selected entity.

A grouping rule can be removed by clicking the [X] button next to the rule.

The query designer will warn the user if the grouping rules are incomplete and/or if the required attributes are not selected into the Group By.

Furthermore, users can filter the aggregate queries using the **Having** conditions. This feature will be enabled only after creating a Group By condition. To access it, open the Group By advance pane. After completing a Group By condition **Having** pane will be visible.

Creating **Having** conditions is similar to creating [Advance Filter rules](#), the only difference being the having conditions are created using aggregation functions.

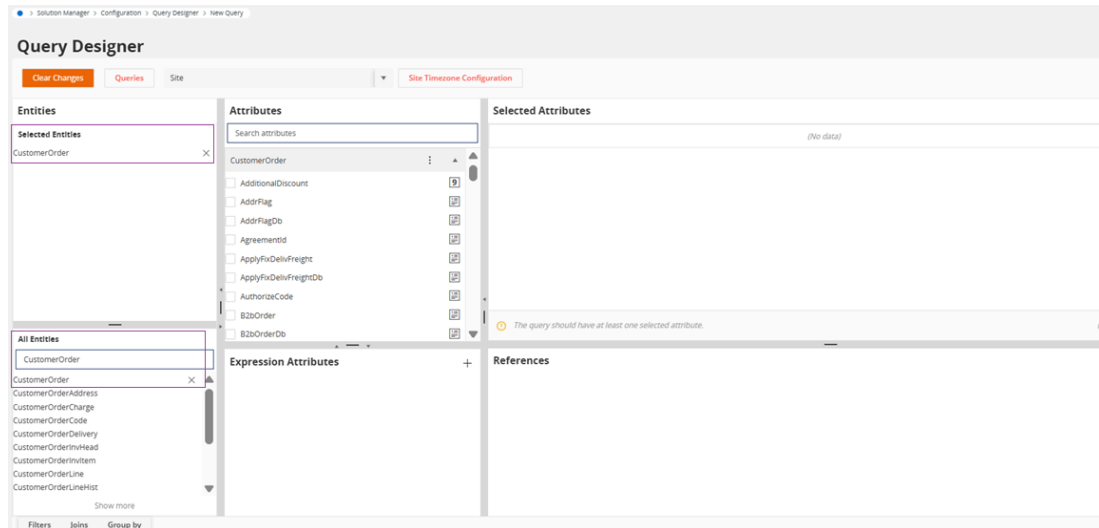
Expression Attributes

The **Expression Attributes** panel enables users to create their own expressions and add them as **Attributes**.

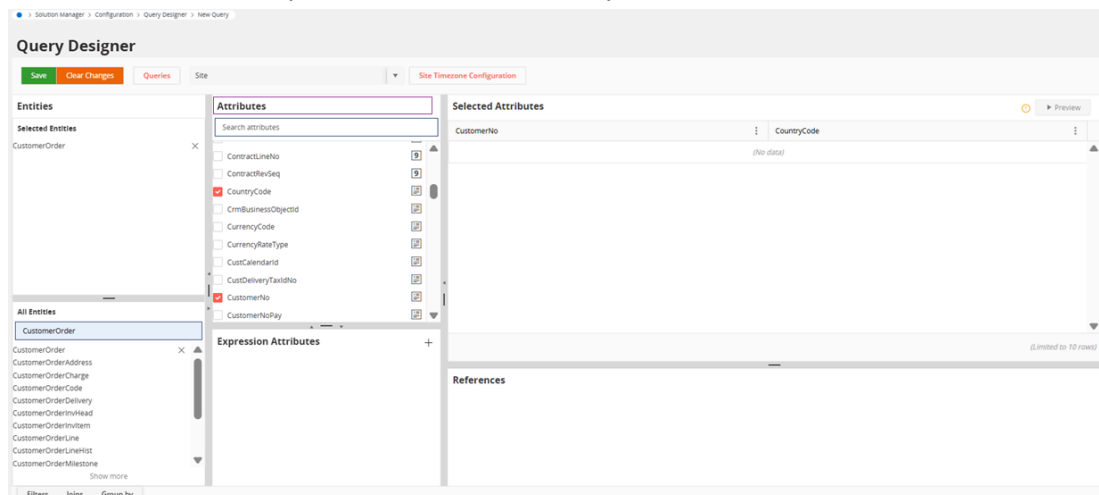
Creating the Expressions

Follow the steps given below to create these expressions:

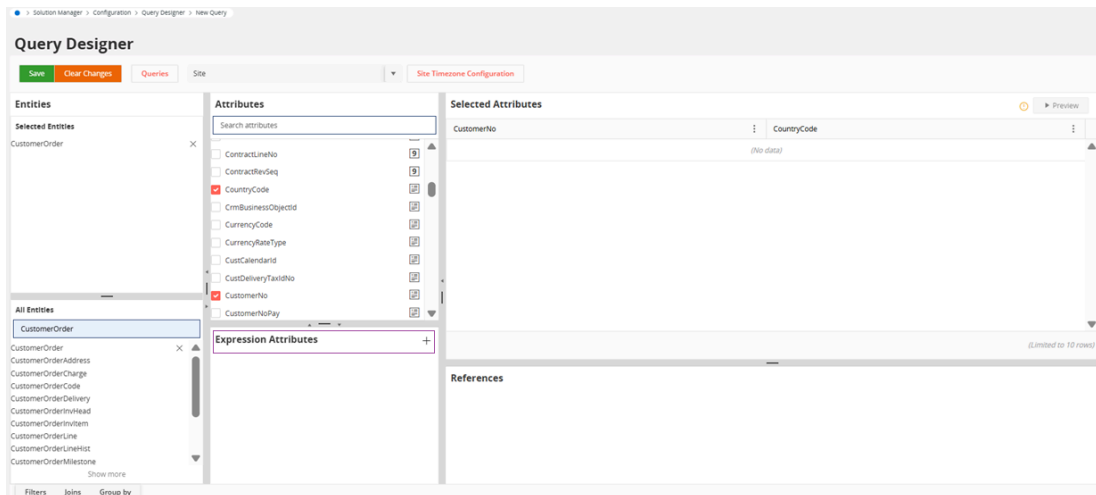
1. Select the entity required from the **All Entities** panel.



2. Select the attributes required from the **Attributes** panel.



3. Click the add icon on the **Expression Attributes** panel. The **Expression Builder** pop-up appears.



4. Enter a suitable name. >Note: The name cannot be modified once saved.

× **Expression Builder**

Configure

Name

Data Type

string
▼

Description

Expression Preview

Builder

Add Block

Add Operator

Save

Cancel

5. Choose the correct **Data Type** of the value that will be returned for the expression you build. Selecting the wrong data type will throw an error.
6. Click the **Add Block** button.

7. Click **Select a block**. A dropdown appears from which you can choose `Static Value`, `Attribute`, `Function`, or `Group` as your block type. These are discussed in detail in the next section.

Note: Once you select a block type, it cannot be modified. You need to remove the incorrect block type manually and then add the required one.

Expression Builder

Configure

Name: Expression-Test

Data Type: string

Description: Concatenate customer number and country code

Expression Preview

Builder

Select a block

- Static Value
- Attribute
- Function
- Group

Save Cancel

8. Once you choose a block type, you can also select the correct block value from the same dropdown which will now have autopopulated the relevant values for the block type you

chose.

Expression Builder

Configure

Name: Expression-Test

Data Type: string

Description: Concatenate customer number and country code

Expression Preview

Builder

Attributes

Search

Customer...

AdditionalDis...

AddrFlag

AddrFlagDb

AgreementId

ApplyFixDeliv...

ApplyFixDeliv...

AuthorizaCode

Save Cancel

9. Click the **Add Operator** button.

10. You can choose the required blocks and operators one after the other to build your expression as shown below. > The blocks are draggable so you can change the order.

×

Expression Builder

Configure

Name

Expression-Test

Data Type

string

Description

Concatenate customer number and country code

Expression Preview

CustomerOrder.CustomerNo || ' ' || CustomerOrder.CountryCode

Builder

CustomerOrder.CustomerNo

||

-

||

CustomerOrder.CountryCode

Add Block

Add Operator

Save

Cancel

11. You can confirm if your expression is correct by looking at the **Expression Preview**.

×

Expression Builder

Configure

Name

Expression-Test

Data Type

string

Description

Concatenate customer number and country code

Expression Preview

CustomerOrder.CustomerNo || ' ' || CustomerOrder.CountryCode

Builder

CustomerOrder.CustomerNo

||

-

||

CustomerOrder.CountryCode

Add Block

Add Operator

Save

Cancel

12. Click **Save**.

The Different Block Types

Static Value

This is used when you need to add static values of different data types to your expression as shown below.

×

Expression Builder

Configure

Name

Static-Test

Data Type

string

Description

Expression Preview

Builder

Static Value

String

Number

Boolean

Date

Date Time

Time

Save

Cancel

×

Expression Builder

Configure

Name

Static-Test

Data Type

string

Description

Expression Preview

'Prepaid' || ' ' || '18/14/2025'

Builder

Prepaid

Static - String

+

18/14/2025

Static - Date

Save

Cancel

Selected Attributes

Preview

CustomerNo	CountryCode	StaticTestAlias
BP10	DE	Prepaid-14-AUG-25
BP8	US	Prepaid-14-AUG-25
BP2	US	Prepaid-14-AUG-25
BP10	DE	Prepaid-14-AUG-25
BP7	US	Prepaid-14-AUG-25
BP1	US	Prepaid-14-AUG-25
BP9	DE	Prepaid-14-AUG-25
BP2	US	Prepaid-14-AUG-25
BP6	SE	Prepaid-14-AUG-25

(Limited to 10 rows)

Attribute

This block type enables you to choose an existing attribute within a selected entity to be used in the expression you want to create.

Once you choose the block type as **Attribute**, the list of attributes within the entity you chose in the **All Entities** panel appears as shown below.

The screenshot shows the **Expression Builder** dialog box. It has a close button (X) in the top left. The title is **Expression Builder**. The dialog is divided into three main sections: **Configure**, **Expression Preview**, and **Builder**.

The **Configure** section contains:

- Name**: A text field with the value "Attribute-Test".
- Data Type**: A dropdown menu with the value "string".
- Description**: An empty text field.

The **Builder** section contains:

- A tab labeled **Attributes** with a dropdown arrow.
- A search bar with the text "Search" and a close button (X).
- A list of attributes: "Customer...", "AdditionalDis...", "AddrFlag", "AddrFlagDb", "AgreementId", "ApplyFixDeliv...", "ApplyFixDeliv...", and "AuthorizeCode".

At the bottom of the dialog are two buttons: **Save** (blue) and **Cancel** (grey).

You can form your expression with the attributes you need as shown below.

×

Expression Builder

Configure

Name

Attribute-Test

Data Type

string

Description

Expression Preview

CustomerOrder.CountryCode || 'OR' || CustomerOrder.LanguageCode

Builder

CustomerOrder.CountryCode

||

OR

CustomerOrder.LanguageCode

Add Block

Add Operator

Selected Attributes

Preview

CustomerNo	CountryCode	LanguageCode	AttributeTestAlias
BP10	DE	en	DEORen
BP8	US	en	USORen
BP2	US	en	USORen
BP10	DE	en	DEORen
BP7	US	en	USORen
BP1	US	en	USORen
BP9	DE	en	DEORen
BP2	US	en	USORen
BP6	SE	sv	SEORsv

(limited to 10 rows)

Save

Cancel

Function

Using this block type, you can add different functions.

×

Expression Builder

Configure

Name

Function-test

Data Type

string

Description

Expression Preview

Builder

Function

Chercher

TO_CHAR

TO_NUMBER

SUBSTR

ROUND

TRUNC

SYSDATE

Save

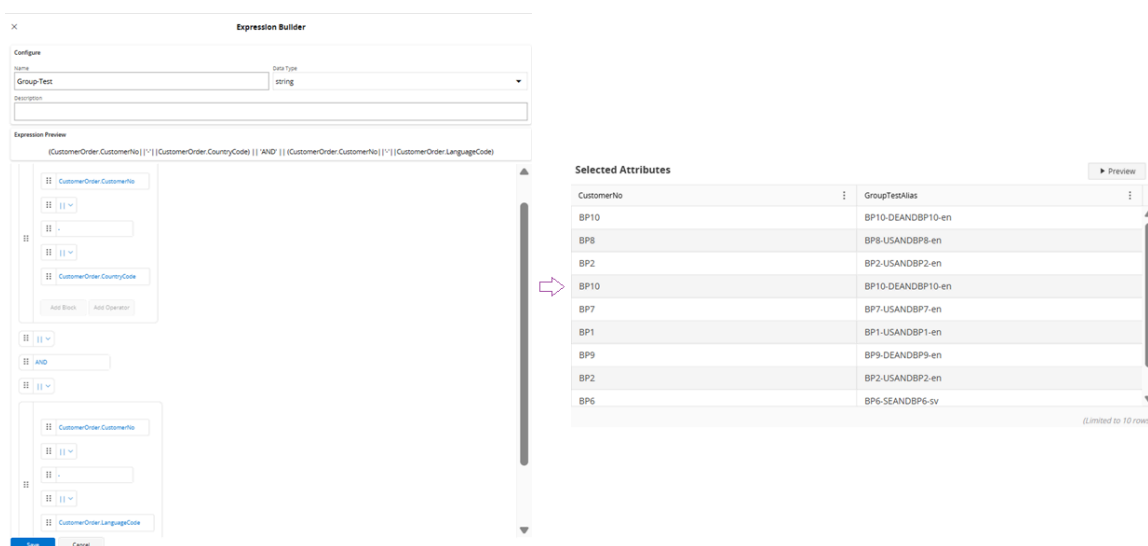
Cancel

Functions can be nested only upto five levels.

Group

The group block can be used in order to group the expression or a portion of it within a set of parentheses.

Similar to functions, groups can also only be applied for upto five levels. Within each group, you can only add up to a total of five operands and operators combined.



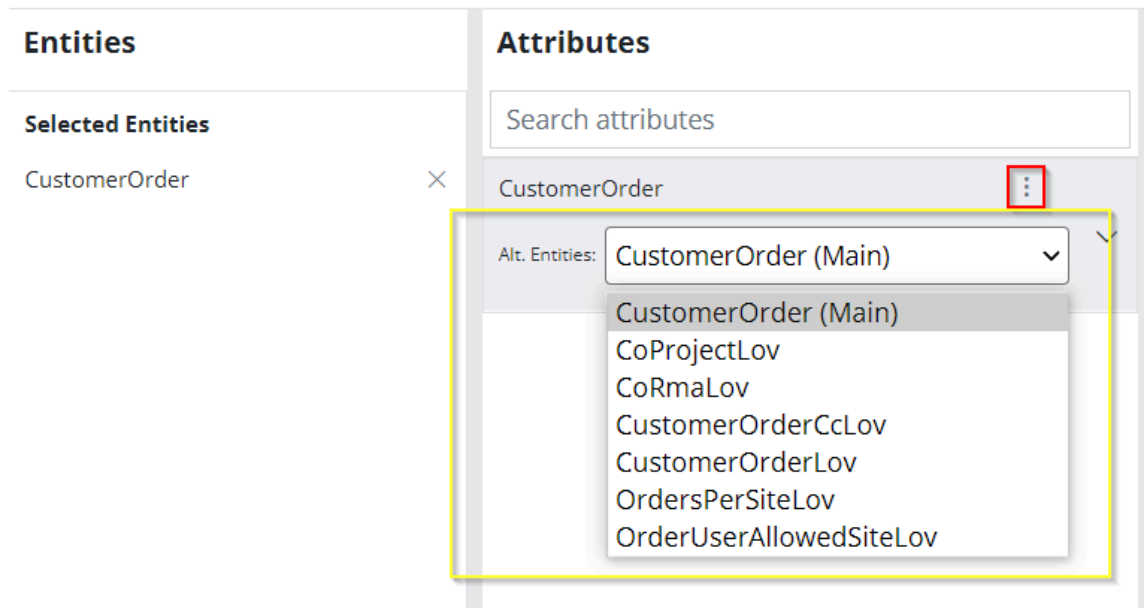
Exposing the Expressions

Follow the instructions to expose the expressions you create as attributes in a query.

1. Select the expression you created from the **Expression Attributes** panel. > You can duplicate the expressions in the **Expression Attributes** panel by clicking on the more options menu shown by three verticle dots adjacent to the expression you choose. Note that you are allowed to modify the name of the duplicate expression only once.
2. Click **Preview** on the **Selected Attributes** panel.

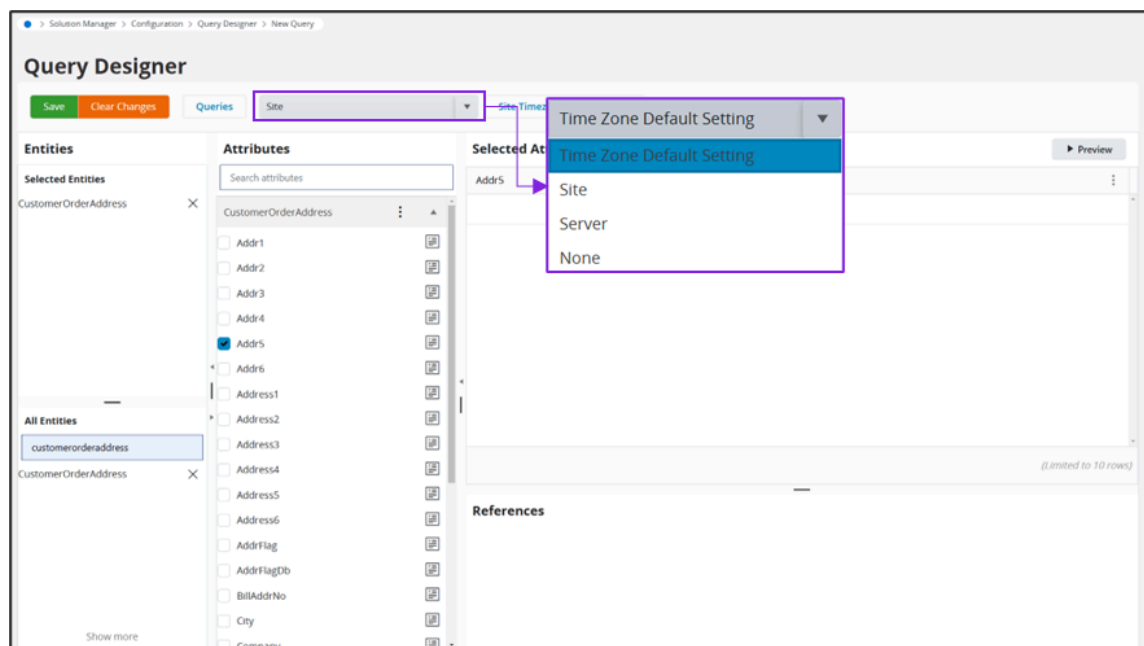
Alternative/Non-base Views

The Query Designer sets the base view by default when an entity is selected. To change to a non-base (alternative) view of a selected entity, click on the menu button next to the entity name in the *Attributes* pane. A list of non-base views will appear. The base view is marked with "(Main)". Select the required non-base view from the list. The attribute list will be updated with the attributes of the selected non-base view.

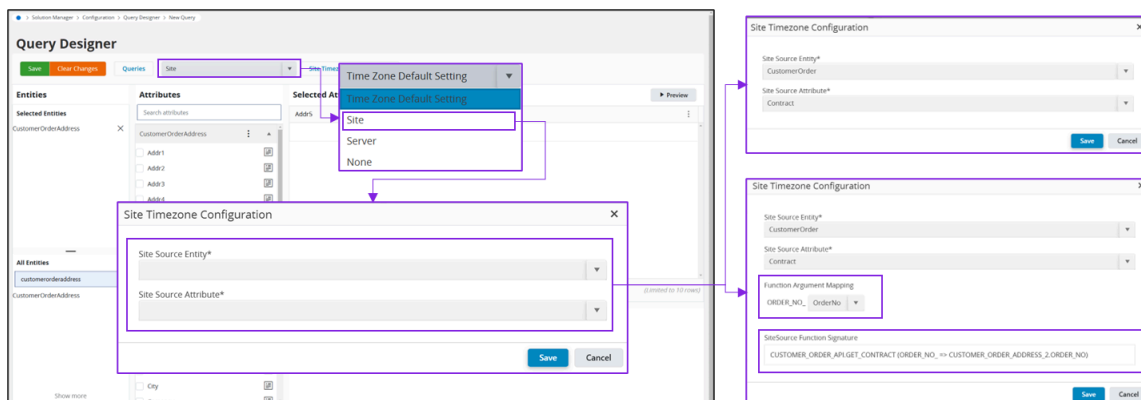


Time zone Selection

In the Query designer, use the **Time Zone Default Setting** dropdown menu, as shown in the image below to annotate a query entity as time zone aware.



- Select **Site** to ensure your query operates with **Site time zone** awareness



- Upon selection, the **Site Time Zone Configuration** pop-up will appear, allowing you to choose the **Entity** and **Attribute**.
- If the entity chosen for the query differs from the entity selected in the **Site Time Zone Configuration** pop-up, additional configuration options will become available. These options allow you to set function parameters in the **Function Argument Mapping** section, while the **Site Source Function Signature** section displays the generated function.
- Select **Server** to ensure your query operates with time zone awareness, allowing it to function based on the selected client time zone.
- Alternatively, choose **None** to disable time zone awareness, causing the query to operate based on the browser time zone.

Note: Consider annotating queries as time zone aware as much as possible, depending on your application's requirements. This practice helps minimize potential confusion when handling date/time values.